Far-field ~~approx.~~ radiation and its angular distribution

$$\vec{A}(r,t) = \frac{\mu_0}{4\pi r}\left(\dot{\vec{d}} + \frac{1}{c}\,\dot{\vec{m}}\times\hat{n} + \frac{1}{6c}\,\ddot{\vec{Q}}\right)$$

$$\vec{B} = \frac{1}{c}\,\dot{\vec{A}}\times\hat{n} \quad , \quad \vec{E} = c\,\vec{B}\times\hat{n}$$

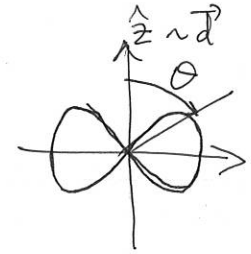
$$I = \frac{c}{\mu_0}\,\overline{\vec{B}^2} \quad \left(= \text{time-averaged intensity}\right) = \frac{1}{T}\int_0^T dT \underbrace{\left(\frac{\vec{E}\times\vec{B}}{\mu_0}\right)}_{\vec{S}}$$

Radiation power emitted into solid angle $d\Omega$

$$\frac{dP}{d\Omega} = I\,R^2\,d\Omega$$

Total emitted power

$$P = \int \frac{dP}{d\Omega}\,d\Omega$$



profile of $\frac{dP}{d\Omega}$

Electric dipole

$$\vec{B}_{dipole} = \frac{\mu_0}{4\pi c r}\,\ddot{\vec{d}}\times\hat{n}$$

$$\vec{E}_{dip} = \frac{\mu_0}{4\pi c r}\left(\ddot{\vec{d}}\times\hat{n}\right)\times\hat{n} = \frac{\mu_0}{4\pi r}\left(\hat{n}\left(\ddot{\vec{d}}\cdot\hat{n}\right) - \ddot{\vec{d}}\right)$$

Charged particle $\vec{d} = q\vec{r} \Rightarrow \ddot{\vec{d}} = q\vec{a}$

$$\frac{dP}{d\Omega} = \frac{c}{\mu_0}\left(\frac{\mu_0}{4\pi c}\right)^2 \overline{\left(\ddot{\vec{d}}\times\hat{n}\right)^2} = \frac{\mu_0}{16\pi^2 c}\,\overline{\left(\ddot{\vec{d}}\times\hat{n}\right)^2} \sim \sin^2\theta$$

$$\left(\ddot{\vec{d}}\times\hat{n}\right)^2 = \left(\ddot{\vec{d}}\times\hat{n}\right)\left(\ddot{\vec{d}}\times\hat{n}\right) = \ddot{\vec{d}}\cdot\left[\hat{n}\times\left(\ddot{\vec{d}}\times\hat{n}\right)\right] = \ddot{\vec{d}}\cdot\left[\ddot{\vec{d}} - \hat{n}\left(\hat{n}\,\ddot{\vec{d}}\right)\right]$$

$$A\cdot B\times C = B\cdot C\times A = C\cdot A\times B$$

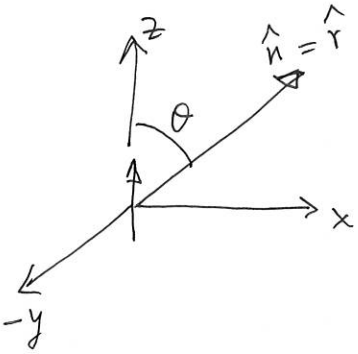
$$\begin{vmatrix} A \\ B \\ C \end{vmatrix} = -\begin{vmatrix} B \\ A \\ C \end{vmatrix} = \begin{vmatrix} B \\ C \\ A \end{vmatrix}$$

$$\left(\ddot{\vec{d}}\times\hat{n}\right)^2 = \ddot{\vec{d}}^{\,2} - \left(\hat{n}\cdot\ddot{\vec{d}}\right)^2 = \ddot{\vec{d}}^{\,2}\left(1-\cos^2\theta\right) = \ddot{\vec{d}}^{\,2}\sin^2\theta$$

choosing $\ddot{\vec{d}}$ along $\hat{z}$ axis

Linearly oscillating dipole:

$\vec{d} = \hat{z}\, d_0 \cos\omega t$



$\hat{n} = \hat{r} = (\sin\theta, 0, \cos\theta)$

in xz-plane

$$\vec{A} = \frac{\mu_0}{4\pi r}\dot{\vec{d}} = -\cancel{\frac{\mu_0\omega}{4\pi r}\vec{d}} = -\frac{\mu_0\omega d_0 \sin\omega t}{4\pi r}\hat{z}$$

$$\vec{B} = \frac{1}{c}\dot{\vec{A}}\times\hat{n} = -\frac{\mu_0\omega^2 d_0 \sin(\omega t_r)}{4\pi c r}\overbrace{\hat{z}\times\hat{n}}^{\hat{y}\sin\theta}$$

retarded time $\boxed{\omega t_r = \omega t - kr}$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ \sin\theta & 0 & \cos\theta \end{vmatrix} = \sin\theta\,\hat{y}.$$

$$\cancel{\vec{B} = -\frac{\mu_0\omega^2|\vec{d}|}{4\pi c r}\hat{y}}$$

$$\vec{E} = c\,\vec{B}\times\hat{n} = \frac{\mu_0\omega^2 d_0\cos(\omega t_r)}{4\pi r}\sin\theta\,(-\cos\theta, 0, \sin\theta).$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & \sin\theta & 0 \\ \sin\theta & 0 & \cos\theta \end{vmatrix} = \hat{x}\sin\theta\cos\theta - \hat{z}\sin^2\theta$$

Therefore: electric field $\vec{E}$ of the wave is linearly polarized in xz $[\vec{d}-\hat{n}]$ plane.
$\vec{B}$ is $\perp$ to that plane (along $\hat{y}$).

The power emitted in the element of a solid angle

$$\frac{dP}{d\Omega} = \frac{c\,\vec{B}^2}{\mu_0}r^2 = \frac{\mu_0}{16\pi^2 c}\omega^4 d_0^2 \sin^2(\omega t_r)\sin^2\theta$$

Time-average over the period:

$$\frac{1}{T}\int_0^T \frac{dP}{d\Omega}\,dt = \int_0^{2\pi} \frac{d(\omega t)}{2\pi}\,\frac{\mu_0 \omega^4 d_0^2}{16\pi^2 c}\,\sin^2(\omega t - kR)\,\sin^2\theta$$

$$k = \frac{\omega}{c}$$

$$T = \frac{2\pi}{\omega}$$

But $\displaystyle\int_0^{2\pi} \frac{du}{2\pi}\,\sin^2(u + \text{const}) = \frac{\pi}{2\pi} = \frac{1}{2}$

$$\Rightarrow \quad \overline{\frac{dP}{d\Omega}} = \frac{\mu_0 \omega^4 d_0^2}{32\pi^2 c}\,\sin^2\theta$$

Total ~~[illegible]~~ power radiated:

$$\overline{P} = \int \overline{\frac{dP}{d\Omega}}\,d\Omega = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta\,\sin\theta\,\overline{\frac{dP}{d\Omega}}$$

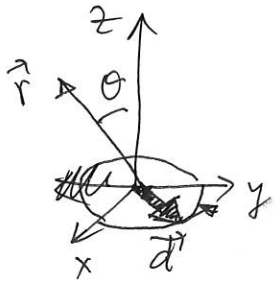
$$\int_0^{\pi} d\theta\,\sin\theta\,\sin^2\theta = -\int_1^{-1} d(\cos\theta)\,(1-\cos^2\theta) =$$

$$= 2\int_0^1 du\,(1-u^2) = 2\left(u - \frac{1}{3}u^3\right)\Big|_0^1 = 2\cdot\frac{2}{3} = \frac{4}{3}$$

$$\overline{P} = \frac{\mu_0 \omega^4 d_0^2}{32\pi^2 c}\,\underbrace{\left(2\pi\cdot\frac{4}{3}\right)}_{\frac{2}{3}\cdot 4\pi}$$

<u>Precessing dipole</u> rotates about $\hat{z}$ axis with angular velocity ω

$$\vec{d} = d_0 \left(\cos\omega t \, \hat{x} + \sin\omega t \, \hat{y} \right)$$



$$\vec{r} = r(\sin\theta, 0, \cos\theta) = r\,\hat{n}$$

$$\vec{E} = c\,\vec{B} \times \hat{n} = \frac{\mu_0}{4\pi r}\left(\ddot{\vec{d}} \times \hat{n}\right) \times \hat{n} = \frac{\mu_0}{4\pi r}\left(\hat{n}\left(\ddot{\vec{d}} \cdot \hat{n}\right) - \ddot{\vec{d}}\right)$$

$$\ddot{\vec{d}} = -\omega^2 \vec{d}$$

$$\vec{d} \cdot \hat{n} = d_0 \sin\theta \cos(\omega t_r)$$

$$\vec{E} = \frac{\mu_0 \omega^2}{4\pi r}\left(\vec{d} - \hat{n}\left(\vec{d} \cdot \hat{n}\right)\right) =$$

$$= \frac{\mu_0 \omega^2 d_0}{4\pi r}\left\{\left(\cos\omega t_r, \sin\omega t_r, 0\right) - \sin\theta \cos\omega t_r \left(\sin\theta, 0, \cos\theta\right)\right\}$$

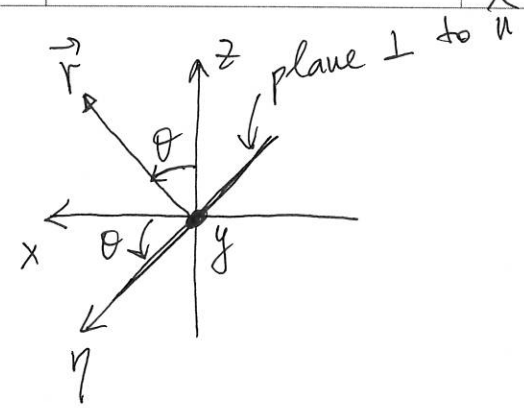
$$= \frac{\mu_0 \omega^2 d_0}{4\pi r}\left(\cos^2\theta \cos\omega t_r, \sin\omega t_r, -\sin\theta\cos\theta \cos\omega t_r\right)$$

Let us project $\vec{E}$ onto the plane perpendicular to $\vec{r} = r\,\hat{n}$.

Observe $\vec{E} \cdot \hat{n} \sim \vec{d} \cdot \hat{n} - \hat{n} \cdot \hat{n}\left(\vec{d} \cdot \hat{n}\right) = 0$

$\Rightarrow$ all field is in the plane $\perp \hat{n}$

This plane is formed by $\hat{\eta} = \hat{x}\cos\theta - \hat{z}\sin\theta$ and $\hat{y}$



$$\Rightarrow \quad E_\eta = \vec{E}\cdot\hat{\eta} = E_x \cos\theta - E_z \sin\theta =$$

$$= \frac{\mu_0 \omega^2 d_0}{4\pi r}\left\{\cos^3\theta \cos(\omega t_r) - (-\sin^2\theta\cos\theta)\cos(\omega t_r)\right\}$$

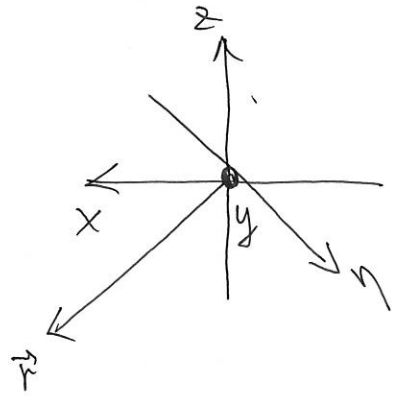
$$\Rightarrow \quad E_\eta = \frac{\mu_0 \omega^2 d_0}{4\pi r}\cos\theta\cos(\omega t_r)$$

$$E_y = \frac{\mu_0 \omega^2 d_0}{4\pi r}\sin(\omega t_r).$$

electric field is elliptically polarized

$E_\eta = \cos\theta\, E_y$

$\vec{E}$ is right-handed for $\theta < \frac{\pi}{2}$ (upper hemisphere)
left-handed for $\frac{\pi}{2} < \theta \le \pi$ (lower hemisphere)



For $\theta = \frac{\pi}{2}$ (equator) $E_\eta = 0$ and $\vec{E}$ is linearly polarized along $\hat{y}$.

$$\frac{dP}{d\Omega} = \frac{1}{\mu_0 c} \vec{E}^2 r^2 = \frac{\mu_0 \omega^4 d_0^2}{16\pi^2 c} \left(\cos^2\theta \cos^2(\omega t_r) + \sin^2(\omega t_r) \right)$$

$$\overline{\frac{dP}{d\Omega}} = \frac{\mu_0 \omega^4 d_0^2}{16\pi^2 c} \cdot \frac{1}{2} \left(1 + \cos^2\theta\right)$$

intensity is greatest along $\theta = 0, \pi$ directions, minimal in equatorial plane $\theta = \pi/2$.

Total intensity

$$\overline{P} = \int \overline{\frac{dP}{d\Omega}} \, d\Omega$$

$$\int d\Omega \left(1 + \cos^2\theta\right) = 2\pi \int_0^\pi d\theta \sin\theta \left(1 + \cos^2\theta\right) = (u = \cos\theta)$$

$$= 2\pi \int_{-1}^{1} du \left(1 + u^2\right) = 4\pi + 4\pi \left.\frac{u^3}{3}\right|_0^1 = 4\pi\left(1 + \frac{1}{3}\right) =$$

$$= 4\pi \cdot \frac{4}{3}$$

$$\overline{P} = \frac{\mu_0 \omega^4 d_0^2}{32\pi^2 c} \underbrace{4\pi \cdot \frac{4}{3}}$$

this is <u>twice</u> the result for $\overline{P}$ of linearly ~~polarized~~ oscillating dipole.

And indeed has to be that way because rotating dipole = <u>two</u> oscillating ones, along $\hat{x}$ and $\hat{y}$ axes, and with $\pi/2$ phase delay.

Magnetic dipole

$$\vec{B} = \frac{1}{c^2}\left(\ddot{\vec{m}} \times \hat{n}\right) \times \hat{n} \left(\frac{\mu_0}{4\pi r}\right) = \frac{-\mu_0}{4\pi r c^2} \left(\hat{n} \times \ddot{\vec{m}}\right) \times \hat{n}$$

$$= -\frac{\mu_0}{4\pi r c^2}\left(\ddot{\vec{m}} - \hat{n}\left(\hat{n} \cdot \ddot{\vec{m}}\right)\right)$$

$$\vec{B}_{md} = \frac{\mu_0}{4\pi r c^2}\left(\hat{n}\left(\hat{n} \cdot \ddot{\vec{m}}\right) - \ddot{\vec{m}}\right)$$

$$\vec{E}_{md} = c\,\vec{B} \times \hat{n} = \frac{\mu_0}{4\pi r c}\left(-\ddot{\vec{m}} \times \hat{n}\right)$$

very similar in form to electric dipole fields

Namely $\vec{d} \to \frac{\vec{m}}{c}$, $\vec{E} \to c\vec{B}$, $\vec{B} \to -\frac{\vec{E}}{c}$

$$\begin{cases} \vec{E}_{dip} = \dfrac{\mu_0}{4\pi r}\left(\hat{n}\left(\ddot{\vec{d}} \cdot \hat{n}\right) - \ddot{\vec{d}}\right) \longrightarrow c\,\vec{B}_{md} = \dfrac{\mu_0}{4\pi r c}\left(\hat{n}\left(\ddot{\vec{m}} \cdot \hat{n}\right) - \ddot{\vec{m}}\right) \\ \vec{B}_{dip} = \dfrac{\mu_0}{4\pi c r}\,\ddot{\vec{d}} \times \hat{n} \longrightarrow -\dfrac{\vec{E}_{md}}{c} = \dfrac{\mu_0}{4\pi c^2 r}\,\ddot{\vec{m}} \times \hat{n} \end{cases}$$

$$\vec{E}_{md} = +\frac{\mu_0}{4\pi c r}\left(\hat{n} \times \ddot{\vec{m}}\right)$$

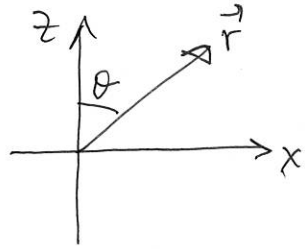
$$I = \frac{c}{\mu_0}\overline{\vec{B}_{dip}^{\,2}} \longrightarrow \frac{c}{\mu_0}\overline{\left(\frac{\vec{E}_{md}}{c}\right)^2} = \frac{c}{\mu_0 c}\left(\frac{\mu_0}{4\pi c r}\right)^2 \overline{\left(\hat{n} \times \ddot{\vec{m}}\right)^2}$$

$$\frac{dP_{md}}{d\Omega} = \frac{\mu_0}{16\pi^2 c^3}\overline{\left(\ddot{\vec{m}} \times \hat{n}\right)^2}$$

Linearly oscillating magnetic dipole

$$\vec{E}_{md} = \frac{\mu_0}{4\pi c r}\, \vec{n} \times \ddot{\vec{m}} = \frac{\mu_0 \omega^2}{4\pi c r}\, \vec{m} \times \hat{n} \qquad \left(\begin{array}{c}\text{perpendicular}\\ \text{to } \vec{m} \text{ and } \hat{n}\end{array}\right)$$

$$\vec{m} = \hat{z}\, m_0 \cos\omega t \Rightarrow \ddot{\vec{m}} = -\omega^2 \hat{z}\, m_0 \cos\omega t = -\omega^2 \vec{m}$$



$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ \sin\theta & 0 & \cos\theta \end{vmatrix} = \sin\theta\, \hat{y}$$

$$\vec{E}_{md} = \frac{\mu_0 \omega^2 m_0 \cos(\omega t_r)}{4\pi c r} \sin\theta\, \hat{y}$$

$$\frac{dP_{md}}{d\Omega} = \frac{\mu_0 \omega^4 m_0^2}{16\pi^2 c^3} \sin^2\theta \cos^2(\omega t_r)$$

$$\frac{d\overline{P}_{md}}{d\Omega} = \frac{\mu_0 \omega^4 m_0^2}{32\pi^2 c^3} \sin^2\theta$$

$$\overline{P}_{md} = \frac{\mu_0 \omega^4 m_0^2}{32\pi^2 c^3} \left(4\pi \cdot \frac{2}{3}\right)$$

but notice additional $\frac{1}{c^2}$ factor relative to $\overline{P}_d$ of electric dipole!

* Similar analysis can be done for precessing mag. dipole
→ one obtains elliptically polarized wave with $E_\eta = E_0 \sin(\omega t_r)$ and $E_y = -E_0 \sin(\omega t_r) \cos\theta$

Electric quadrupole

$$\vec{A} = \frac{\mu_0}{24\pi c r} \ddot{\vec{Q}} \quad , \quad \ddot{\vec{Q}} = \ddot{\overleftrightarrow{Q}}\,\hat{n}$$

i.e. $Q_\alpha = Q_{\alpha\beta}\, n_\beta$

$$\vec{B} = \frac{\mu_0}{24\pi c^2 r} \dddot{\vec{Q}} \times \hat{n}$$

$$\frac{dP}{d\Omega} = \frac{c}{\mu_0}\left(\frac{\mu_0}{24\pi c^2}\right)^2 \overline{\left(\dddot{\vec{Q}} \times \hat{n}\right)^2} = \frac{\mu_0}{24^2\pi^2 c^3}\overline{\left[\dddot{\vec{Q}}^2 - \left(\hat{n}\cdot\dddot{\vec{Q}}\right)^2\right]}$$

$$\dddot{\vec{Q}}^2 = \sum_\alpha \dddot{Q}_\alpha^{\,2} = \sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\, n_\beta\, \dddot{Q}_{\alpha\gamma}\, n_\gamma$$

$$\hat{n}\cdot\dddot{\vec{Q}} = \sum_\alpha \dddot{Q}_\alpha\, n_\alpha = \sum_{\alpha\beta} \dddot{Q}_{\alpha\beta}\, n_\alpha n_\beta$$

$$\frac{dP}{d\Omega} = \frac{\mu_0}{576\pi^2 c^3}\overline{\left\{\sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\alpha\gamma}\, n_\beta n_\gamma - \sum_{\alpha\beta\gamma\delta} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\gamma\delta}\, n_\alpha n_\beta n_\gamma n_\delta\right\}}$$

Total emitted power

$$P = \int \frac{dP}{d\Omega}\, d\Omega \quad \Rightarrow \text{ we need to average}$$

$$\underbrace{\frac{1}{4\pi}\int d\Omega\, n_\beta n_\gamma}_{\frac{1}{3}} \quad \text{and} \quad \underbrace{\int \frac{d\Omega}{4\pi}\, n_\alpha n_\beta n_\gamma n_\delta}_{\frac{1}{15}}$$

(see next page)

Averaging over component product of $\hat{n}$:

$$\frac{1}{4\pi}\int d\Omega\, n_\beta n_\gamma = K_1 \delta_{\beta\gamma}$$ (no angular dependence after averaging)

$$\Rightarrow K_1 = \frac{1}{4\pi}\int d\Omega\, n_3^2 = \frac{1}{4\pi}\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \cos^2\theta =$$

$$= \frac{1}{2}\int_{-1}^{1} d(\cos\theta)\cos^2\theta = \frac{1}{2}\cdot 2 \cdot \int_0^1 dx\, x^2 = \frac{1}{3}.$$

$$\overline{n_\beta n_\gamma} = \frac{1}{3}\delta_{\beta\gamma}$$

$$\overline{n_\alpha n_\beta n_\gamma n_\delta} = \frac{1}{4\pi}\int d\Omega\, n_\alpha n_\beta n_\gamma n_\delta =$$

$$= K_2\left(\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}\right)$$

$$\Rightarrow 3K_2 = \frac{1}{4\pi}\int d\Omega\, \overline{n_1^4} = \frac{1}{2}\int_0^{\pi} d\theta \sin\theta \cos^4\theta = \frac{1}{2}\cdot 2\cdot\int_0^1 dx\, x^4 = \frac{1}{5}$$

$$K_2 = \frac{1}{15}.$$

$\Rightarrow$ the total power by quadrupole

$$I = \frac{\mu_0}{576\pi^2 c^3}\left\{\sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\alpha\gamma}\,\overline{n_\beta n_\gamma} - \right.$$

$$\left. - \sum_{\alpha\beta\gamma\delta} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\gamma\delta}\,\overline{n_\alpha n_\beta n_\gamma n_\delta}\right\} =$$

$$= \frac{\mu_0}{576\pi^2 c^3}\left\{\sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\alpha\gamma}\,\frac{4\pi}{3}\delta_{\beta\gamma} - \right.$$

$$\left. - \sum_{\alpha\beta\gamma\delta} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\gamma\delta}\,\frac{4\pi}{15}\left(\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}\right)\right\}$$

$$\{\ \} = \sum_{\alpha\beta} \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\alpha\beta}\,\frac{4\pi}{3} - \frac{4\pi}{15}\sum_{\alpha\beta}\left(\underbrace{\dddot{Q}_{\alpha\alpha}}_{=0}\,\underbrace{\dddot{Q}_{\beta\beta}}_{=0}\right.$$

$$\left. + \dddot{Q}_{\alpha\beta}\,\dddot{Q}_{\alpha\beta} + \dddot{Q}_{\alpha\beta}\,\underbrace{\dddot{Q}_{\beta\alpha}}_{=\dddot{Q}_{\alpha\beta}}\right) =$$

$$= \sum_{\alpha\beta}\left(\dddot{Q}_{\alpha\beta}\right)^2\left(\frac{4\pi}{3} - \frac{4\pi}{15}\cdot 2\right)$$

$$\frac{4\pi}{15}(5-2) = \frac{4\pi}{5}$$

$$\Rightarrow I = \frac{\mu_0}{720\pi c^3}\sum_{\alpha\beta}\left(\dddot{Q}_{\alpha\beta}\right)^2 .$$

But $\dddot{\vec{Q}} \Rightarrow \quad Q_\alpha = Q_{\alpha\beta}\, n_\beta$

vector with components

$$\Rightarrow \quad \dddot{\vec{Q}}^{\,2} = \sum_\alpha Q_\alpha^2 = \sum_{\alpha\beta\gamma} (Q_{\alpha\beta}\, n_\beta)(Q_{\alpha\gamma}\, n_\gamma) =$$

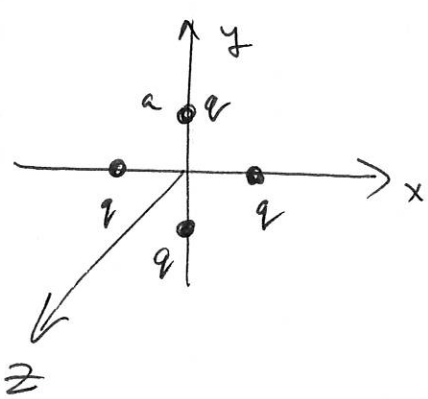
$$= \sum_{\alpha\beta\gamma} Q_{\alpha\beta}\, Q_{\alpha\gamma}\, n_\beta\, n_\gamma$$

$$\dddot{\vec{Q}} \cdot \hat{n} = \sum_{\alpha\beta} Q_{\alpha\beta}\, n_\beta\, n_\alpha$$

$$\frac{dI}{d\Omega} = \frac{\mu_0}{576\pi^2 c^3} \left\{ \sum_{\alpha\beta\gamma} \dddot{Q}_{\alpha\beta}\, \dddot{Q}_{\alpha\gamma}\, n_\beta\, n_\gamma - \right.$$

$$\left. - \sum_{\alpha\beta\gamma\delta} \dddot{Q}_{\alpha\beta}\, n_\beta\, n_\alpha\, \dddot{Q}_{\gamma\delta}\, n_\gamma\, n_\delta \right\}$$

<u>Example</u> Consider quadrupole with spheroidal symmetry about z axis:



quadrupole moment

$$Q = 2qa^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$\vec{Q} = 2qa^2 (n_x, n_y, -2n_z) =$$

vector

$$= Q(t)\,(n_x, n_y, -2n_z)$$

$$Q_{xx} = Q_{yy} = -\frac{1}{2} Q_{zz} \quad , \vec{n} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$$

$$\left.\begin{aligned} Q_x &= Q_{x\alpha} n_\alpha = Q_{xx} n_x \\ Q_z &= Q_{zz} n_z = -2 Q_{xx} n_z \end{aligned}\right\} \vec{Q} = Q_{xx}(n_x, n_y, -2n_z)$$

$$\left(\dddot{\vec{Q}}\right)^2 = \left(\dddot{Q}_{xx}\right)^2 (n_x^2 + n_y^2 + 4n_z^2) = \left(\dddot{Q}_{xx}\right)^2 (\sin^2\theta + 4\cos^2\theta)$$

$$\left(\dddot{\vec{Q}} \cdot \hat{n}\right)^2 = \left(\dddot{Q}_{xx}\right)^2 (n_x^2 + n_y^2 - 2n_z^2)^2 = \left(\dddot{Q}_{xx}\right)^2 (\sin^2\theta - 2\cos^2\theta)^2$$

$$\cancel{\sin^2\theta + 4\cos^2\theta}$$

$$\sin^2\theta + 4\cos^2\theta - (\sin^2\theta - 2\cos^2\theta)^2 = 1 + 3\cos^2\theta -$$

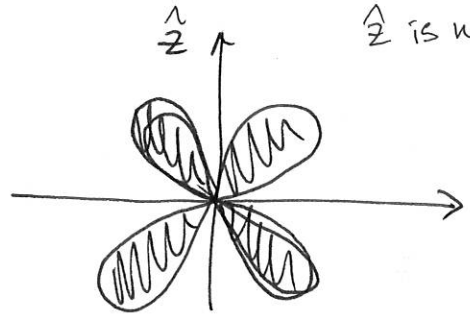
$$- (1 - 3\cos^2\theta)^2 = 1 + 3\cos^2\theta - 1 + 6\cos^2\theta - 9\cos^4\theta$$

$$= 9\cos^2\theta(1 - \cos^2\theta) = 9\cos^2\theta\sin^2\theta$$

$\Rightarrow$ for such a ~~symme~~ axially symmetric quadrupole:

$$\frac{dI}{d\Omega} = \frac{\mu_0}{576\pi^2 c^3} \left(\dddot{Q}_{xx}\right)^2 9\cos^2\theta\sin^2\theta$$

$$= \frac{\mu_0}{64\pi^2 c^3} \left(\dddot{Q}_{xx}\right)^2 \cos^2\theta\sin^2\theta$$



$\hat{z}$ is normal to the plane of quadrupole